Assignment 1

Each question is worth 20 points.

1. Suppose we have a sample of N pairs (x_i, y_i) drawn i.i.d. from the distribution characterized as follows:

$$
\begin{aligned}
&x_i \sim h(x), \quad \text{the design density} \\
&y_i = f(x_i) + \epsilon_i, \quad \text{f is the regression function} \\
&\epsilon_i \sim (0, \sigma^2), \quad \text{mean zero, variance } \sigma^2
\end{aligned}
$$

We construct an estimator for f *linear* in the y_i,

$$\hat{f}(x_0) = \sum_{i=1}^{N} \ell_i(x_0; \mathcal{X}) y_i,$$

where the weights $\ell_i(x_0, \mathcal{X})$ do not depend on the y_i, but do depend on the entire training sequence of x_i, denoted here by $\mathcal{X}$.

(1a) Show that linear regression and k-nearest-neighbor regression are members of this class of estimators. Describe explicitly the weights $\ell_i(x_0; \mathcal{X})$ in each of these cases.

(1b) Decompose the conditional mean-squared error

$$\mathbb{E}_{\mathcal{Y}|\mathcal{X}}(f(x_0) - \hat{f}(x_0))^2$$

into a conditional squared bias and a conditional variance component. Like $\mathcal{X}$, $\mathcal{Y}$ represents the entire training sequence of y_i.

(1c) Decompose the (unconditional) mean-squared error

$$\mathbb{E}_{\mathcal{Y},\mathcal{X}}(f(x_0) - \hat{f}(x_0))^2$$

into a squared bias and a variance component.

(1d) Establish a relationship between the squared biases and variances in the above two cases.

2. Suppose that $N \geq 2$, and we observe N pairs $\{x_i, y_i\}_{i=1}^N$, with $a < x_1 < \cdots < x_N < b$. Consider the penalized least squares problem

$$\min_f \left[\sum_{i=1}^N (y_i - f(x_i))^2 + \lambda \int_a^b f''(t)^2 dt \right].$$

Argue that the minimizer must be a cubic spline with knots at each of x_i.

3. Suppose that we are given $N = 2048$ noisy samples of a function f,

$$y_i = f(t_i) + z_i, \quad i = 1, \ldots, N,$$

with $t_i = (i-1)/N$, z_i iid $N(0, \sigma^2)$. Our goal is to estimate the vector $\mathbf{f} = (f(t_i))_{i=1}^N$ with small mean square error. That is, to find an estimate $\hat{\mathbf{f}}$ depending on $y_1, \ldots, y_N$ with small risk $R(\hat{\mathbf{f}}, \mathbf{f}) = N^{-1}\mathbb{E}\|\hat{\mathbf{f}} - \mathbf{f}\|_2^2$.

We consider two true functions for f:

(a) Bumps: a sum of bumps $\sum_{j=1}^{11} h_j K((t - t_j)/w_j)$ with locations

$$(t_j) = (0.1, 0.13, 0.15, 0.23, 0.25, 0.40, 0.44, 0.65, 0.76, 0.78, 0.81),$$

heights

$$(h_j) = (4, 5, 3, 4, 5, 4.2, 2.1, 4.3, 3.1, 5.1, 4.2),$$

width

$$(w_j) = (0.005, 0.005, 0.006, 0.01, 0.01, 0.03, 0.01, 0.01, 0.005, 0.008, 0.005),$$

and the function $K(t) = 1/(1 + |t|)^4$.

(b) Doppler: the variable-frequency signal

$$f(t) = \sqrt{t(1-t)}\sin(\frac{2.1\pi}{(t+0.05)}).$$

The noise is independent $N(0,1)$ with $\sigma = 1$. We rescale the function to have signal-to-noise ratio $SD(f)/\sigma = 7$. The wavelet fit has been reported by Donoho and Johnstone (1995)— "Adapting to Unknown Smoothness via Wavelet Shrinkage".

Implement two other regression methods and report the root mean square over 20 replications of the root loss $N^{-1/2}\|\hat{\mathbf{f}} - \mathbf{f}\|_2$ (not its square). Explain clearly why you can or cannot beat the wavelet method (Table 2 in Donoho and Johnstone (1995)). Python codes need to be submitted for reproducibility. Your score is partially determined by how far is your result from Donoho's result.